Dariusz A. Wójcik

Ph.D.

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Profile

Applied Mathematician and early-career Space Physicist with expertise in turbulence, stochastic processes, probability theory, and space data analysis. Experienced in working with the NASA's *Magnetospheric Multiscale* (*MMS*) mission satellite data, statistical modeling, numerical analysis, and interdisciplinary research bridging mathematics and space physics, as well as finances and insurances. Proficient in applying advanced mathematical frameworks such as *Markov processes*, *Fokker-Planck* and *Langevin* dynamics, and *multifractal* analysis to investigate energy transfer mechanisms in space plasma turbulence across kinetic scales. Author of various peer-reviewed publications in the leading physical and astrophysical journals.

Education

Mar 2022 – Sep 2025 **Ph.D. in Space Physics (Exact and Natural Sciences)**, *Space Research Centre, Polish Academy of Sciences*, Warsaw, Poland
Thesis: *Analysis of Markov Processes in Space Environment*. With Honors (***Cum Laude***).
Highlight: Distinction for progress in the doctoral work and for publication achievements.

Oct 2019 – Jan 2022 **M.Sc. in Mathematics (Insurance & Finance)**, *Warsaw University of Technology*, Warsaw, Poland
Thesis: *Cybersecurity Insurance Modeling*.
Highlight: Rector's Scholarship.

Oct 2019 – Dec 2021 **M.Sc. in Mathematics (Economy)**, *Cardinal Stefan Wyszyński University*, Warsaw, Poland
Thesis: *Stability Analysis of the Generalized Lorenz System*.
Highlight: Rector's Scholarship.

Oct 2016 – Jul 2019 **B.Sc. in Mathematics (Finance)**, *Cardinal Stefan Wyszyński University*, Warsaw, Poland
Thesis: *Mathematical Modeling of the Heartbeat*. With Honors (***Cum Laude***),
Highlight: Rector's Scholarship.

Research Positions

Sep 2025 – present **Research Fellow (Space Physicist)**, *Space Research Centre, Polish Academy of Sciences*, Warsaw, Poland
Task: Analysis of cosmic plasma in the magnetosphere, heliosphere, and astrophysical applications.

Research Interests

Space Plasma; Turbulence; Solar Wind; Magnetosphere; Magnetic Reconnection; Stochastic Processes; Statistical Modeling; Dynamics; Differential Equations; Cybersecurity Modeling; Non-life Insurances.

Publications

Main Peer-Reviewed Papers

Feb 2026 **W. M. Macek and D. Wójcik**, *Multifractal Spectrum Observed in the Universe Distribution of Galaxies*, Chaos 1, 36 (2): 023129. https://doi.org/10.1063/5.0289242.

Jan 2026 **W. M. Macek and D. Wójcik**, *Fractal Nature of Galaxy Clustering in the Updated CfA Redshift Catalog*, Scientific Reports, 16, 6181, https://doi.org/10.1038/s41598-026-36013-3.

Sep 2025 **D. Wójcik and W. M. Macek**, *Searching for Universality of Turbulence in the Earth's Magnetosphere*, Journal of Geophysical Research: Space Physics, 130(10), e2025JA034020, https://doi:10.1029/2025JA034020.

Aug 2024 **D. Wójcik and W. M. Macek**, *Testing for Markovian Character of Transfer of Fluctuations in Solar Wind Turbulence on Kinetic Scales*, Physical Review E, 110(2), 025203, https://doi.org/10.1103/PhysRevE.110.025203.

Sep 2023 **W. M. Macek and D. Wójcik**, *Statistical Analysis of Stochastic Magnetic Fluctuations in Space Plasma Based on the MMS Mission*, Monthly Notices of the Royal Astronomical Society, 526(4), 5779–5790, https://doi.org/10.1093/mnras/stad2584.

Feb 2023 **W. M. Macek, D. Wójcik, J. L. Burch**, *Magnetospheric Multiscale Observations of Markov Turbulence on Kinetic Scales*, The Astrophysical Journal, 943(2):152, https://doi.org/10.1029/2025JA034020 (*listed by* LASP Colorado: https://lasp.colorado.edu/galaxy/spaces/mms).

Other Papers

Feb 2026 **A. Bandari**, *Does the distribution of galaxies in the universe follow a large-scale fractal pattern?*, AIP Scilight, 2026 (8):081104, ***Research highlight of***: **W. M. Macek and D. Wójcik** (2026), https://doi.org/10.1063/10.0042454.

Oct 2025 **R. Ringuette, M. Hughes et al., incl. D. Wójcik**, *Open Science for Missions and Observatories: Practical Choices to Shape Your Observing Adventure* (In Review), https://tinyurl.com/OpenScienceMissions.

Feb 2024 **S. Dorfman, E. Lichko et al., incl. D. Wójcik**, *Next Generation Solar Wind Facility for Discovery Plasma Science*, Fusion Energy Sciences Advisory Committee *White Paper*. https://tinyurl.com/fesac24.

Research Projects

Jan 2022 – Jul 2025 **Turbulence and Magnetic Reconnection in Earth's Space Environment**, *National Science Centre (NCN)*, Kraków, Poland, *Doctoral Scholarship Holder*
Title: Analysis of high-resolution *MMS* mission data to study turbulence and reconnection in the solar wind and magnetosphere.

Conferences & Presentations

Aug 2026 **COSPAR Scientific Assembly 2026**, Florence, Italy,
Title: *From Homogeneity to Multifractality in the Distribution of Distant Galaxies*.

May 2026 **European Geosciences Union (EGU) 2026**, Vienna, Austria,
Title: *Multifractal Analysis of the Large-Scale Galaxy Distribution*.

Dec 2025 **American Geophysical Union (AGU) 2025**, New Orleans, LA, USA,
Title: *Markovian Universality and Local Asymmetry of Turbulent Cascades in Earth's Magnetosphere: Multiscale MMS Study*.

Oct 2025 **Cluster – Plasma Observatory 2025 (INVITED)**, Paris, France,
Title: *Universality of Markovian Turbulence in Earth's Magnetosphere Based on MMS Data*.

May 2025 **European Geosciences Union (EGU) 2025**, Vienna, Austria,
Title: *Testing For Universality of Markov Solar Wind Turbulence at the Earth's Magnetosphere on Kinetic Scales Based on the MMS Mission*.

Dec 2024 **American Geophysical Union (AGU) 2024**, Washington DC, USA,
Title 1: *Multifractal Spectrum Observed in the Distribution of Galaxies*
Title 2: *MMS Observations of Markov Turbulence in the Magnetosheath on Kinetic Scales*.

May 2024 **EMCEI 2024**, Marrakesh, Morocco,
Title: *Exploring Solar Wind Turbulence: MMS Mission Comparative Data Analysis*.

Apr 2024 **European Geosciences Union (EGU) 2024**, Vienna, Austria,
Title: *Probing Small Scale Solar Wind Turbulence: Markovian Analysis and Scale Interactions from Inertial to Kinetic Regimes*.

Jul 2023 **SigmaPhi Statistical Physics 2023**, Crete, Greece,
Title: *MMS Observations of Kappa Distributions in the Magnetosheath on Small Scales*.

Jun 2023 **CHAOS International Conference 2023**, Crete, Greece,
Title: *Fokker-Planck Statistical Analysis of Turbulence on Kinetic Scales.*

May 2023 **American Geophysical Union (AGU): Chapman - Alfvén Waves 2023**, Berlin, Germany,
Title: *Markov Analysis of Magnetic Turbulence on Kinetic Scales Based on MMS Data.*

Apr 2023 **European Geosciences Union (EGU) 2023**, Vienna, Austria,
Title: *Comparative MMS Analysis of Markov Turbulence in the Magnetosheath on Kinetic Scales.*

Technical Skills

Programming R, Python, SQL, Mathematica, IDL

Mathematics Statistics, Probability Theory, Data Analysis, Differential Equations, Dynamical Systems, Algebra

Tools MATLAB, TeX, Streamlit, Git, SPEDAS, PlasmaPy, AstroPy, SpacePy, CDFlib

Soft Skills Problem-solving, Growth mindset, Critical thinking, Adaptability, Communication, Attention to Detail

Experience

2025 – present **Peer-reviewer** for *Scientific Reports* – Springer Nature, and *Entropy* / *Atmosphere* MDPI articles.

Jan 2023 – Jan 2027 **Member** of the European Geosciences Union (EGU), American Geophysical Union (AGU), and Japan Geoscience Union (JpGU).

Jun 2024 **Certificates**: Data Science: Building Machine Learning Models, Harvard University, edX.

Languages

Polish Native

English Fluent

German Basic

Japanese Basic

Professional References

- Formal **letters of recommendation** are available upon request from the senior researchers at *Southwest Research Institute* (*SwRI*) *Texas, USA*; *KTH Royal Institute of Technology, Stockholm, Sweden*; *Warsaw University of Technology, Poland*; *Polish Academy of Sciences*.
- Informal professional endorsements from researchers at NASA Goddard and Princeton University.

Outreach articles

- *Is the universe like a cauliflower? Researchers from the Space Research Center PAS sought answers.*
  Project Pulsar.
- *Mathematics behind the turbulent space weather: why Markov processes matter for modern technology?*
  GeoPlanet School CAMK Warsaw.
- *Polish-American team uncovers new details of space turbulence*
  Science in Poland.
- *The mechanism responsible for magnetic turbulence in space around Earth*
  Cardinal Stefan Wyszynski University Warsaw.